

Smart Distribution Systems

Forecasting Assignment

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Forecasting Assignment

- **Create a neural network to forecast day-ahead electricity prices**

- As good as possible

- **Evaluation is based on the Mean Squared Error:**
$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2.$$

- **Forecast the hourly Belpex prices for the period from 12/05/2026 at 00:00 to 14/05/2026 at 23:00.**

- **Available data** (will be provided and uploaded to Toledo on the morning of **12 May 2026**):

- Day-Ahead Price BE (from 01/01/2021 00:00 to 11/05/2026 23:00)
- Load Forecast FR (from 01/01/2021 00:00 to 11/05/2026 23:00)
- Generation Forecast FR (from 01/01/2021 00:00 to 11/05/2026 23:00)
- Day-Ahead Price Switzerland (from 01/01/2021 00:00 to 11/05/2026 23:00)
- Wind Generation Forecast BE (from 01/01/2021 00:00 to 11/05/2026 23:00)
- Solar Generation Forecast BE (from 01/01/2021 00:00 to 11/05/2026 23:00)
- Load Forecast BE (from 01/01/2021 00:00 to 11/05/2026 23:00)
- **You can work together with others (2-3 people per group)**

Forecasting Assignment

- **Create a predictions.csv file with 72 rows, each row should contain the forecast for that hour (3 predicted days).**
 - => 1 column with 72 rows
 - MSE calculation is automated -> If your prediction is not in the correct format, it will not be evaluated.
 - **Do not include a column label nor an index column, the only column of your csv should be your forecasted values.**
- **Write a report of 3 to 5 pages (including plots) with the results of your work!**
 - Add plots, performance metrics (table), explanation of feature selection and network architecture selection.
- The **updated and final dataset** will be provided and uploaded to Toledo on the morning of **12 May 2026**.
- Until then, train your model on the **Full_Dataset.csv** file provided for Exercise Session 2, which has been updated as of 25 February 2026.
- Deadline: **13 May 2026, at 11:59 pm**
 - Please zip your files: NN model, predictions.csv, and report, and name the zip file using your student number(s).
 - Upload the zip file to the dedicated Toledo page for the assignment.